Requesting a Freshness Nonce for Attestation Evidence in CSRs

draft-ietf-lamps-attestation-freshness

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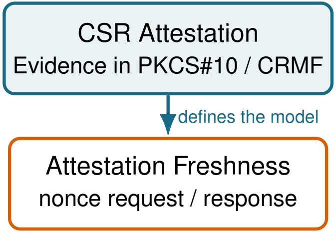
Why the larger update now?

A direct dependency

This document defines freshness for Evidence carried using `draft-ietf-lamps-csr-attestation`.

The CSR Attestation draft made significant progress earlier this year. That provided a firmer basis for:

- the architecture and terminology,
- the relation between a nonce and Evidence, and
- support for multiple Attesters and Evidence statements.



The -08 changes align the two drafts.

Changes from -05 to -08

-05

- "Conveying a nonce" separately in CMP, EST, and CMC
- Arrays / sequences of nonce requests
- `type` and free-form `hint` tied to a Verifier
- Protocol-specific structures evolved independently

$\Longrightarrow$

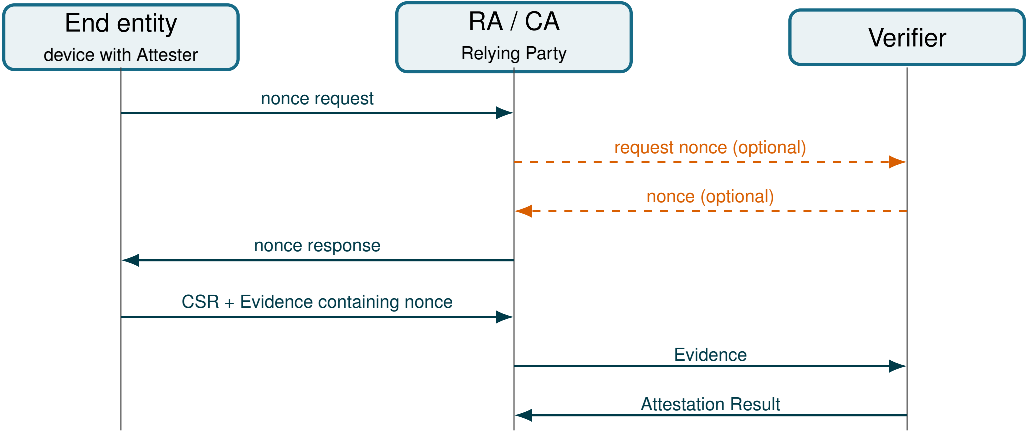
-08

- One common architecture and message model
- One nonce request and one nonce response
- Typed, extensible `reqInfo` / `respInfo`
- Consistent ASN.1 and CDDL representations
- Protocol bindings are separate from message semantics

Scope

Request a nonce before enrollment. Then, include it inside the Evidence carried by the subsequent CSR.

Architecture



In scope: the interface between the End Entity and the Relying Party (RA/CA).
Out of scope: the interface between the Relying Party and the Verifier.

Freshness: nonce vs. claim data

Nonce-based freshness

- The RP or Verifier provides an unpredictable challenge; the Attester binds it to the Claims in protected Evidence.
- A match proves that the Evidence was produced after the challenge.
- An extra exchange and typically per-nonce state are required.

In scope: This document specifies the nonce exchange.

Freshness data in Claims

- The time when the Claim values were collected by the Attesting Environment is, in general, unknown.
- Claim values may have changed between their collection and the production of the Evidence.
- Freshly signed Evidence does not by itself prove that individual Claim values were freshly collected.

Out of scope: This document neither defines nor evaluates freshness data in Claims.

Extensibility

NonceRequest

len optional, 8–64 bytes

reqTypeInfo optional

- `type`: OID identifying the syntax
- `reqInfo`: type-specific input

NonceResponse

nonce required, 0 or 8–64 bytes

expiry optional, seconds

respTypeInfo optional

- `type`: OID identifying the syntax
- `respInfo`: type-specific output

ASN.1 for CMP / CMC **CDDL for EST JSON / CBOR**

`reqInfo` and `respInfo` are defined by external, OID-identified profiles.

Protocol bindings are aligned

Protocol	Transport mechanism	Encoding	Notable -08 detail
CMP	`genm` / `genp`	ASN.1	New information types; HTTP `getnonce` and CoAP `nonce` paths
EST over HTTPS	`GET` or `POST` on `/nonce`	JSON	Dedicated `application/est-attestation-freshness+json` media type
EST over CoAPs	`GET` or `POST` on `/nonce`	CBOR	CoAP response codes and a dedicated CBOR media type / Content-Format
CMC	Controls in Full PKI Request	ASN.1	Separate request / response control OIDs and explicit status handling

Same semantics across CMP, EST, and CMC; only the binding changes.

EST is no longer an open design point

-05

- HTTPS + generic `application/json`
- JSON arrays and index-based correlation
- Absolute timestamp for `expiry`
- EST-over-CoAP / CBOR left as an open issue

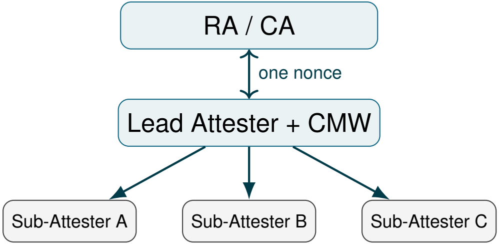
-08

- HTTPS/JSON and secure CoAP/CBOR specified
- CDDL defines both representations
- One object; no positional correlation
- `expiry` consistently means validity in seconds
- Explicit success and error response codes
- Dedicated JSON and CBOR media types

GET versus POST

`GET` when no optional request content is needed; `POST` when `len` or type-specific request information is conveyed.

Multiple Attesters



Conceptual Message Wrapper (CMW) combines
the individual Evidence statements

- -05 allowed arrays of requests and responses.
- -08 uses one nonce and one `respInfo`.
- A lead Attester distributes the nonce to sub-Attesters.
- CMW can carry multiple Evidence statements and, if needed, multiple type-specific information elements.

This follows the composite Attester model instead of inventing protocol-specific correlation.

Operational and Security Requirements

Nonce handling

- At least 64 bits of entropy
- Unique and never reused
- 8–64 byte wire representation
- No arbitrary truncation or padding
- Any transformation must be deterministic and profile-defined
- For Evidence conveyance in CSR, the nonce is inside the Evidence payload

Deployment considerations

- Keep request, response, and CSR in the same PKI management context
- Bound outstanding state, lifetime, and request rate
- Treat type-specific information as potentially privacy-sensitive
- Clearly separate freshness of Evidence from semantics of other claims

New registrations

CMP and CMC

- CMP URI path segments
- CMP request / response information-type OIDs
- CMC request / response control OIDs
- ASN.1 module OID

EST

- EST JSON and CBOR media types
- CoAP Content-Format
- Dedicated types replace generic `application/json`
- HTTPS/JSON and secure CoAP/CBOR are both covered

Result

-08 specifies the registry surface needed to implement all three protocol bindings; the remaining numeric assignments are marked TBD.

TPM 2.0 Example with JSON for EST over HTTP

1. End entity advertises its capabilities


```
{
  "len": 32,
  "reqTypeInfo": {
    "type": "1.2.3.4.5",
    "reqInfo": {
      "certificate-name": ["aik-1", "aik-2"],
      "supported-hash-algo": ["TPM_ALG_SHA256"]
    }
  }
}
```


Two Attestation Keys are available; SHA-256 is supported.

2. RA/CA selects challenge parameters


```
{
  "nonce": "<32-byte nonce, base64url>",
  "expiry": 600,
  "respTypeInfo": {
    "type": "1.2.3.4.6",
    "respInfo": {
      "tpm20-pcr-selection": [{
        "tpm20-hash-algo": "TPM_ALG_SHA256",
        "pcr-index": [0, 1, 2, 3]
      }],
      "certificate-name": "aik-1"
    }
  }
}
```


Use `aik-1`; quote PCRs 0–3 from the SHA-256 bank.

The Attester creates a TPM quote using these selections and the nonce. The quote is carried later as Evidence in the CSR—not in `respInfo`.

Informative example only; the OIDs are placeholders and no TPM profile is defined.

Summary and next steps

- -08 is an architectural alignment with CSR Attestation
- one extensible nonce exchange
- consistent protocol bindings
- explicit support for composite-Attester use cases
- open-source implementation: `https://github.com/Guiliano99/RAT-E2E`.

One open issue

Establish an IANA registry for EST path segments; addressed by PR #38.